

Teacher Perception and Practices of the Next Generation Science Standards (NGSS): Insights from a New England Study

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Abstract: This qualitative study explores how secondary science teachers understand and implement two core dimensions of the Next Generation Science Standards (NGSS): Science and Engineering Practices (SEPs) and Crosscutting Concepts (CCCs). Drawing on semi-structured interviews with ten teachers from schools serving economically diverse communities in a New England state, the analysis reveals distinct orientations toward SEPs. These orientations range from identity-based and assessment-aligned approaches to emerging understanding through professional learning. In contrast, CCCs appeared less intentionally addressed, with most teachers using them as implicit outcomes rather than explicit instructional goals. A few teachers, however, used CCCs to promote interdisciplinary connections and support comparative reasoning. These findings highlight variation in teachers' enactment of NGSS dimensions and the need for ongoing professional learning that supports integration of SEPs and CCCs. The study contributes to efforts to bridge the gap between policy vision and classroom realities in three-dimensional science teaching and assessment.

Introduction

Science education in the United States is changing. In 2013, states adopted the Next Generation Science Standards (NGSS) to align classroom learning with a new national vision (NGSS Lead States, 2013). That vision comes from *the Framework for K–12 Science Education*, which calls for three-dimensional learning that blends disciplinary core ideas, science and engineering practices, and crosscutting concepts (National Research Council [NRC], 2012). The goal is coherence. Students should learn science by doing science and by seeing ideas that connect across topics (NRC, 2012; NGSS Lead States, 2013). Professional groups echo this shift and describe three-dimensional teaching and learning as the expected baseline for modern science instruction (National Science Teaching Association, 2018). Research has shown that when students engage in Science and Engineering Practices (SEPs) and Crosscutting Concepts (CCCs) alongside disciplinary ideas, they develop deeper conceptual understanding, stronger reasoning skills, and greater motivation to learn science (He et al., 2024; Nordine & Lee, 2023; Sherwood & Benedict-Chambers, 2025).

Yet a decade on, research continues to show that the implementation of the NGSS vision remains uneven. Classrooms vary widely in how deeply teachers integrate the three dimensions or how consistently they connect them in instruction (He et al., 2024). Studies reveal that while teachers often adopt the language of NGSS, they struggle to coordinate the disciplinary ideas, practices, and crosscutting concepts in meaningful ways (Nordine & Lee, 2023; NRC, 2012). A recent analysis showed that implementation of three-dimensional science in secondary schools remains lower than expected nationwide (Smith, 2020) and may be influenced by the complex interplay of factors such as access to professional development, curriculum resources, teacher knowledge, and administrative support (Haag & Megowan-Romanowicz, 2021; He et al., 2024). Together, these studies show that three-dimensional learning often remains *aspirational*, shaped by contextual constraints, uneven professional learning, and differing interpretations of what it means to teach science as practice.

Within this uneven landscape, Science and Engineering Practices have received the most attention. Teachers commonly report that engaging students in SEPs helps make learning active and authentic (He et al., 2024; Krajcik & Merritt, 2012). While studies have demonstrated the promise of SEPs for improving student engagement and understanding (Kang et al., 2019), research also highlights persistent challenges in enacting them effectively. In a large-scale study across more than 1,200 schools, Banilower et al. (2018) found that science teachers at both elementary and secondary levels struggled to integrate SEPs meaningfully into instruction. These findings suggest that while SEPs are central to the NGSS vision, consistent implementation remains difficult.

Crosscutting Concepts, on the other hand, are the least visible dimension of NGSS in classroom instruction (Nordine & Lee, 2023; Sherwood & Benedict-Chambers, 2025). Although CCCs are intended to promote coherence and transfer across topics, some evidence shows that teachers rarely mention or assess them directly (Nordine & Lee, 2023). Even when teachers acknowledge CCCs conceptually, few report using them to guide instruction or assessment design (He et al., 2024). This limited engagement suggests that CCCs, while valued in theory, remain difficult to translate into practice.

Thus, understanding how teachers interpret, practice, and assess both SEPs and CCCs is critical for explaining the persistent gap between the NGSS framework and its classroom realities. Building on this context, this study explores *how secondary science teachers understand and enact the three-dimensional vision of the NGSS in their daily instruction*. Mainly, it focuses on how teachers interpret and use Science and Engineering Practices (SEPs) and Crosscutting Concepts (CCCs) in teaching and assessment.

NGSS Framework

The present study is grounded in the three-dimensional learning model proposed by *the Framework for K–12 Science Education* (NRC, 2012) and operationalized through the Next Generation Science Standards (NGSS Lead States, 2013). The NGSS defines science learning as the integration of disciplinary knowledge (via DCIs), epistemic practices (via SEPs), and conceptual connections (via CCCs) across scientific domains. This framework reconceptualizes science education as active sensemaking—where students explain, model, and reason about phenomena using the tools and habits of scientists (Krajcik & Merritt, 2012; Penuel & Reiser, 2018). Rather than treating the dimensions as separate strands, the framework emphasizes their interaction: SEPs provide ways of knowing, CCCs connect ideas across contexts, and DCIs anchor understanding in key disciplinary ideas. Together, they aim to develop students who can apply scientific reasoning flexibly and coherently (Laverty et al., 2016; NRC, 2012).

Using NGSS as a theoretical lens allows this study to analyze how teachers conceptualize and enact these intertwined dimensions within real classroom systems. Prior work has shown that the success of three-dimensional instruction depends heavily on teachers' *interpretations* and *contextual* constraints (He et al., 2024; Lee & Arias, 2024). Teachers' understanding of SEPs and CCCs, in particular, serves as a window into how the NGSS vision is translated into daily teaching and assessment. Framing the study through the NGSS therefore provides both a conceptual map and an evaluative lens for examining how teachers make sense of and implement the two key dimensions of three-dimensional learning.

Methodology

This qualitative study uses an interpretivist approach (Creswell & Poth, 2018; Schwandt, 2000) to explore secondary science teachers' understanding and enactment of the three-dimensional vision of the Next Generation Science Standards (NGSS), focusing on SEPs and CCCs. The research draws on data originally collected to examine how secondary science teachers adapted instruction during the COVID-19 lockdown in Spring 2020 (Raj, 2022). Subsequent analyses of this dataset have examined equity (Raj et al., 2024), differentiated instruction (Raj et al., 2025a), and teacher resilience (Raj et al., 2025b). Building on this work, the present study draws on the same interviews to explore teachers' reflections on NGSS-aligned practices, offering insight into how they conceptualize and implement SEPs and CCCs.

Participants included ten secondary science teachers (six male, four female) from a New England state, representing high-, middle-, and low-socioeconomic contexts. Teachers were recruited using a combination of convenience and purposeful sampling. Semi-structured Zoom interviews were conducted during the summer of 2020, lasting between 45 minutes and two hours. Conversations focused on teachers' instructional adaptations, assessment approaches, and perspectives on the three dimensions of science learning. Although participants taught in varied socioeconomic settings, the present analysis did not reveal systematic differences, so this paper centers on shared conceptions and practices.

Data were analyzed using Braun and Clarke's (2006) thematic analysis framework. The process began with inductive coding to identify salient ideas and patterns, followed by deductive refinement guided by the NGSS dimensions—particularly SEPs and CCCs—which served as analytic lenses for interpretation. This iterative approach allowed for an integrated understanding of teacher sensemaking, capturing both emergent and theory-informed insights. The findings below highlight how teachers in our study interpret and apply SEPs and CCCs within their classroom practice.

Findings

Spectrum of Science and Engineering Practices

While most teachers mentioned that Disciplinary Core Ideas (DCIs) were relatively easier to teach and assess, their engagement with Science and Engineering Practices (SEPs) ranged from full integration to general awareness. For some teachers, SEPs are not simply pedagogical strategies but an *identity* to cultivate. Hank, for instance, viewed his classroom as a space where students gradually *become* scientists. He explained, "*I tell my*

students that I do expect them to be functioning physicists at the end of the year... I spend a lot of time focusing on defining questions, creating experiments to answer those questions...creating models... and the critical thinking skills involved in assessing those models.” His expectation that students “be functioning physicists” reveals how he treats SEPs as habits of mind rather than an isolated academic skill. Similarly, Heather echoed this framing when she explained that students “ask questions...design a way to look for answers...we use CER method where they make a Claim...give Evidence...and then the Reasoning behind that.” Her emphasis on the CER method reflects an effort to help students practice structured scientific reasoning. Together, these examples illustrate teachers’ mindset and practices when integrating SEPs into their instruction.

At the same time, several teachers approached SEPs through structured systems of assessment and accountability. For instance, Liam explained, “All of my grades are assigned to a science and engineering practice...when I make an assignment...I’m a little more intentional about what types of questions those assessments have, because I want to make sure that I can really justify that it does apply very clearly to one of those practices.” His reflection shows a deliberate effort to align every task with a specific practice, ensuring clarity between teaching and evaluation of SEPs. Similarly, Mike described how his school’s graduation system formally embeds SEPs into accountability structures: “We just assigned the science skills to [school name] high school graduation standards. So, students have to show proficiency in those skills to graduate.” This example highlights a school-level commitment where SEPs are not only taught but also institutionalized as criteria for demonstrating competence. Together, these accounts illustrate classrooms and school systems that have established measurable and transparent indicators of SEPs-related learning.

While some teachers demonstrated confidence in integrating SEPs, others were still developing a deeper understanding through professional learning. Mia shared, “I’ve come a long way with it, because I’ve had lots of opportunities with professional development...a lot of this is that discovery piece and allowing students to discover a lot of these things on their own, and then applying it in that assessment.” Her reflection highlights how sustained professional learning can strengthen teachers’ confidence and shape their teaching practices with SEPs.

By contrast, Lydia described navigating structural barriers that limited her ability to fully implement SEPs. She explained, “I’m just getting, I’m just delving into this [NGSS]... I wanted our power standards to be the practices... but that was not what was desired by administration... my assessments are going to be tied to the core content...without looking specifically at the crosscutting concepts or SEPs...I do post them all in my classroom... I do let students know what the crosscutting concepts are... and I do bring them up when I introduce the course content standards.” Her comments reveal both awareness and limitation as she values SEPs but works within a system that prioritizes content coverage over integrated practices.

Across teachers, these differences reflected three orientations: identity-driven, system-driven, and evolving through professional learning. In contrast, some teachers, like Lydia, faced institutional constraints that limited their ability to move beyond content-focused instruction. Together, these perspectives show that teachers’ engagement with SEPs varies widely, shaped by their experiences and school contexts.

Limited Crosscutting Concepts in Practice

Crosscutting Concepts (CCCs) were the least discussed and least intentionally assessed dimension of NGSS among our participants. Most teachers did not even bring up CCCs when reflecting on their approach to NGSS during remote instruction. They focused instead on how they handled core content and attempted different ways to include SEPs in their teaching and assessment. Considering that CCCs were not the direct topic of investigation, we did not probe deeply during interviews. However, some evidence suggests that teachers’ attention to CCCs was minimal and often implicit.

For instance, Heather admitted, “Yeah, I don’t think I have that intentionally assessed... Let me think. Yeah... I definitely didn’t consciously think of ways to assess that [CCCs] differently.” Her hesitation captures the uncertainty many teachers felt about where CCCs fit within their instruction. Heather’s reflection reveals that she does not reject the idea of CCCs but had not yet found clear ways to integrate or assess them within her existing routines. This suggests that while teachers were aware of CCCs conceptually, they were still exploring how to connect them meaningfully to instruction and assessment.

Hank, who otherwise demonstrated a highly intentional approach to SEPs above, expressed a similar stance. He reflected, “I do not focus much on the crosscutting concepts before or after remote instruction... I use the content standards to inform me and then make sure they apply and are clearly applicable to SEPs.” His statement shows that even among teachers with well-developed assessment systems for SEPs, CCCs often fall outside their instructional priorities. Both Heather and Hank’s reflections suggest that CCCs are acknowledged but not always translated into specific instructional or assessment practices. While most participants could name

the seven crosscutting concepts—patterns, cause and effect, scale, systems, energy and matter, structure and function, and stability and change—few reported weaving them systematically into lessons.

We did find two disconfirming examples where teachers described using CCCs in more intentional or natural ways. Hazel explained, *“crosscutting concepts is not really a big problem to assess remotely, because it can often be done with a writing piece or being able to look at similarities and differences... my friends in environmental science designed their entire end of the year around looking at these kinds of crosscutting concepts.”* Her comment reveals a clear awareness of CCCs as a thinking framework that encourages comparison, and connection. Similarly, another teacher, Luke, described a more interdisciplinary perspective when talking about CCCs. He explained, *“We did with our freshman group... we have two interdisciplinary units that we do with Social Studies and English. One of them... was about the changing nature of our community... from the science end of it, we look at human impact... of our activities on our communities.”* Luke’s reflection shows how CCCs can connect ideas across subjects and act as an opportunity for inter-subject collaboration among teachers, allowing students to see how scientific and social perspectives intersect.

Overall, teachers’ engagement with crosscutting concepts mirrored what we observed with Science and Engineering Practices, showing a spectrum of understanding shaped by context and support. Some teachers interpreted CCCs as implicit patterns that emerge during inquiry, while others actively integrated them through writing or interdisciplinary projects. However, for many, CCCs remained less defined and less intentionally embedded compared to SEPs. These variations suggest that while both dimensions share the potential to deepen reasoning and connection making, CCCs are often the least implemented part of three-dimensional learning.

Discussion and Implications

For SEPs

Findings from this study highlight a clear range in how teachers understood and used SEPs. Some, like Hank and Heather, viewed SEPs as identity-building tools, encouraging students to see themselves as scientists through activities such as modeling, designing experiments, and using evidence-based reasoning. Others, such as Liam and Mike, approached SEPs through structured assessment systems that linked grades and graduation standards directly to specific practices. A few, like Mia, described themselves as still learning to use SEPs intentionally through professional development. In contrast, Lydia highlighted the institutional barriers that limited this process, noting that her school’s emphasis on core content often left little room to prioritize SEPs. These variation mirrors earlier research showing that while teachers value SEPs for promoting active and meaningful science learning, their classroom use often depends on experience, training, and context (He et al., 2024; Krajcik & Merritt, 2012).

These results reinforce the need for systemic supports that help teachers engage meaningfully with SEPs. As He et al. (2024) found, strong implementation depends on access to professional learning, collaboration, and reform-oriented instructional resources. Our findings similarly suggest that even motivated teachers struggle to sustain SEP-focused instruction without institutional alignment and shared structures for growth. Educational institutions must prioritize comprehensive support systems that include rich materials, institutional support, and sustained professional development (Brand, 2020; Luft et al., 2022).

For CCCs

Teachers in this study seldom described teaching or assessing crosscutting concepts (CCCs) intentionally, a finding consistent with research trends (Nordine & Lee, 2023). While most participants were familiar with the CCCs, many found them difficult to apply in everyday instruction. Teachers such as Hank acknowledged that they did not focus on CCCs directly, often prioritizing content coverage or SEPs. This pattern reflects earlier work showing that CCCs are often treated as implicit outcomes rather than planned goals (Fick & Arias, 2022).

At the same time, several teachers showed that CCCs can be integrated naturally when supported by written assignments or collaboration. Hazel described using writing and comparison tasks to help students recognize relationships and patterns, while Luke shared interdisciplinary lessons linking environmental science with social studies and English. Their approaches align with Nordine and Lee’s (2023) description of CCCs as tools for connecting ideas across contexts and supporting sensemaking. Lee and Arias (2024) similarly argue that implementing CCCs requires *complex* and *multifaceted* knowledge, combining content understanding with pedagogical skill. Fick and Arias (2022) note, thoughtful use of CCCs can also promote equity by making science learning more connected and accessible. Taken together, these insights suggest that teachers are aware of CCCs’ potential but need sustained professional learning to help them apply these concepts intentionally in classroom.

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